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The Editors-in-Chief
Journal of Cheminformatics
BioMed Central
London, UK

Dear Editors,

We are pleased to submit our manuscript, **“Quantum-Inspired Molecular Representations for African Natural Product Chemical Space: Topological Fingerprints, Tensor Network Embeddings, and Quantum Kernel Scores”**, for consideration as an **Original Research Article** in the *Journal of Cheminformatics*.

This work introduces three novel molecular descriptors—Topological Fingerprint (TFP), Tensor Network Embedding (TNE), and Quantum Kernel Score (QKS)—benchmarked against classical fingerprints on 19,849 African NP-derived candidates. We demonstrate that persistent homology resolves the scaffold paradox in generative AI (92.6% Tanimoto novelty with 69.3% scaffold recovery), TNE achieves $15.6\times$ compression on a standard workstation, and quantum kernels match classical baselines after proper hyperparameter tuning (QKS AUC 0.751 vs RBF 0.701, $p = 0.088$, ns). We report honest negative results: classical ECFP4 remains superior for primary screening (AUC 0.868 vs hybrid 0.842, $p = 0.111$).

All data and code are deposited on Zenodo and GitHub under the MIT licence. This work is original, not under consideration elsewhere, and all authors have approved the submission.

We thank you for your consideration.

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Suggested reviewers:

- Prof. Bing Wang (topological methods in drug discovery)
- Dr. Guillaume Godin (RDKit, natural product cheminformatics)
- Prof. Gisbert Schneider (generative molecular design)